

Computing Sums w/ Summation Notation

T.A. Hatice Obuz (hatice.obuz@yasar.edu.tr)

Summation Properties

$$\sum_{i=1}^{n} c \,.\, a_i \;= c \,.\, \sum_{i=1}^{n} a_i \qquad (1.1)$$

$$\sum_{i=1}^{n} a_i \pm b_i \;= \sum_{i=1}^{n} a_i \;\pm \sum_{i=1}^{n} b_i \qquad (1.2)$$

Summation Formulas

$$\sum_{i=1}^{n} 1 = n \qquad \sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6} \qquad \sum_{i=1}^{n} i^3 = \frac{n^2(n+1)^2}{4}$$

Example

Express

$$\frac{1}{3} - \frac{1}{9} + \frac{1}{27} - \frac{1}{81} + \frac{1}{243}$$

in summation notation.

Solution

$\frac{1}{3} - \frac{1}{9} + \frac{1}{27} - \frac{1}{81} + \frac{1}{243}$ Observe, $3^1 = 3, 3^2 = 9, 3^3 = 27, 3^4 = 81, \ and\ 3^5 = 243.$

For odd i (1,3,5) the fraction is positive and for even i (2,4) the fraction is negative.

So, the equivalent statement is:

$$\sum_{i=1}^{5} \frac{(-1)^{(i+1)}}{3^i}$$

Example

Express

$$\frac{2}{3}+\frac{4}{9}+\frac{8}{27}+\frac{16}{81}+\frac{32}{243}$$

in summation notation.

Solution

Observe, $(\frac{2}{3})^1 = \frac{2}{3}$, $(\frac{2}{3})^2 = \frac{4}{9}$, $(\frac{2}{3})^3 = \frac{8}{27}$, $(\frac{2}{3})^4 = \frac{16}{81}$, $(\frac{2}{3})^5 = \frac{32}{243}$

So, the equivalent statement is:

$$\sum_{i=1}^{5} (\frac{2}{3})^i$$

Example

Express

$$\sum_{i=1}^{6} \sqrt{2}i \qquad \text{and} \qquad \sum_{k=1}^{7} \frac{(-1)^{k+1}}{k^2}$$

in expanded notation.

Solution

$$\sum_{i=1}^{6} \sqrt{2}i = \sqrt{2} + 2\sqrt{2} + 3\sqrt{2} + 4\sqrt{2} + 5\sqrt{2} + 6\sqrt{2}$$

$$\sum_{k=1}^{7} \frac{(-1)^{k+1}}{k^2} = 1 - \frac{1}{4} + \frac{1}{9} - \frac{1}{16} + \frac{1}{25} - \frac{1}{36} + \frac{1}{49}$$

Example

Evaluate

$$\sum_{i=1}^{10}(4i-3)$$

RECALL

$$\sum_{i=1}^{n} 1 = n \qquad \sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

Solution

$$\sum_{i=1}^{10}(4i-3) = 4\sum_{i=1}^{10} i - 3\sum_{i=1}^{10} 1$$

$$= 4\ x\ \frac{10\ x\ 11}{2} - 3\ x\ 10$$

$$= 220 - 30 = 190$$

Example

Evaluate

$$\sum_{i=1}^{6}(i^3 - i^2)$$

Solution

RECALL

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6} \qquad \sum_{i=1}^{n} i^3 = \frac{n^2(n+1)^2}{4}$$

$$\sum_{i=1}^{6}(i^3 - i^2) = \sum_{i=1}^{6} i^3 - \sum_{i=1}^{6} i^2$$

$$= \frac{n^2(n+1)^2}{4} - \frac{n\,(n+1)(2n+1)}{6}$$

$$= \frac{36\ x\ 49}{4} - \frac{6\ x\ 7\ x\ 13}{6}$$

$$= 350$$

Example

Evaluate

$$\lim_{n\to\infty} \frac{1^3 + 2^3 + \cdots + n^3}{n^4}$$

RECALL

$$\sum_{i=1}^{n} i^3 = \frac{n^2(n+1)^2}{4}$$

Solution

$$1^3 + 2^3 + \cdots + n^3 = \sum_{i=1}^{n} i^3$$

$$= \frac{n^2(n+1)^2}{4}$$

$$\lim_{n\to\infty} \frac{n^2(n+1)^2}{4\,n^4} = \frac{n^4 + 2n^3 + 1}{4n^4} = \frac{d}{dn}\left(\frac{n^4+2n^3+1}{4n^4}\right) = \frac{1}{4}$$

THE END